Questions

Q1.

Figure 20 shows a satellite orbiting the Earth.

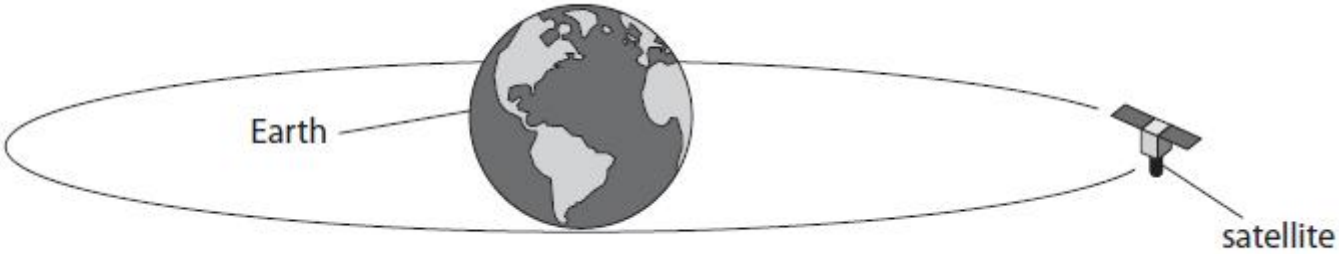


Figure 20

(i) State the name of the force that keeps the satellite in orbit around the Earth.

(1)

.....

(ii) Draw an arrow on Figure 20 to show the direction of the force acting on the satellite, that keeps the satellite in orbit around the Earth.

Label this arrow 'F'.

(1)

(Total for question = 2 marks)

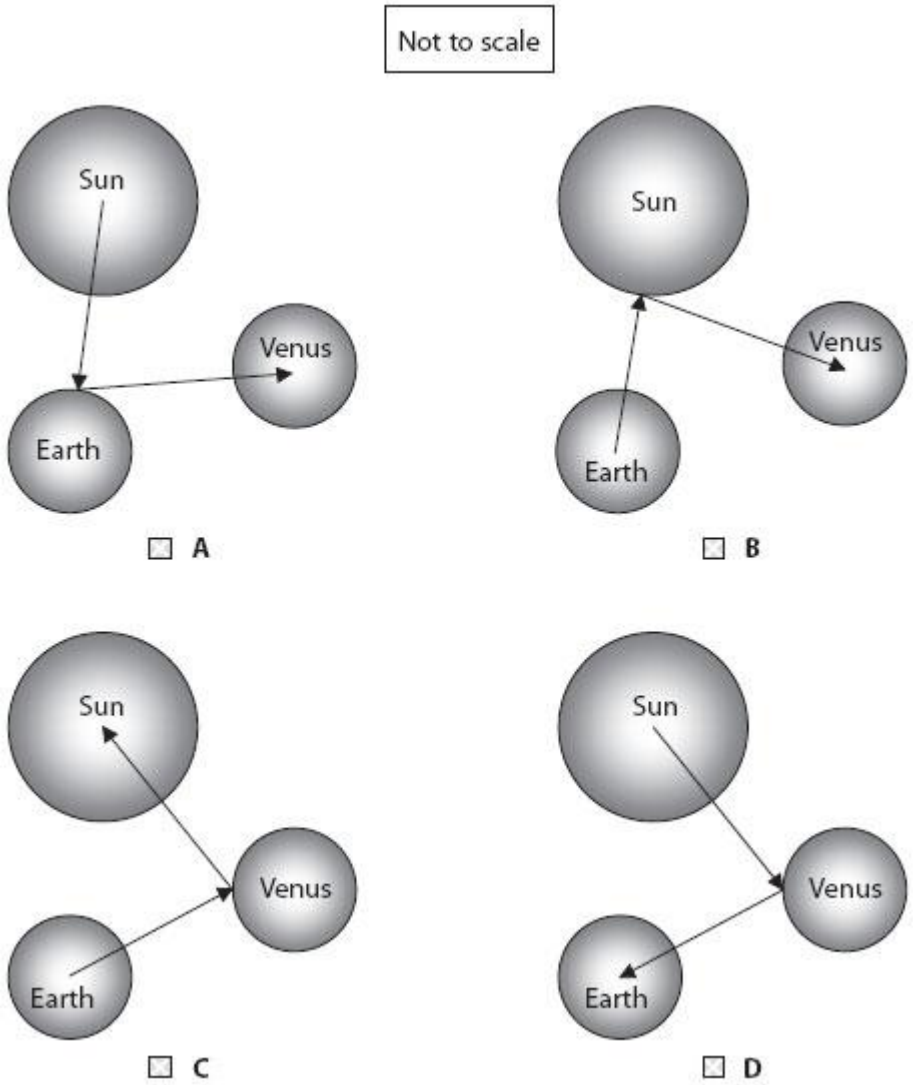
Q2.

Galileo was one of the first scientists to use a telescope to study Venus.

(i) Which of these diagrams best shows how light waves enable us to see Venus?

Put a cross (☒) in the box next to your answer.

(1)



(ii) Use words from the box to complete the sentences.

(2)

asteroids	comets	geocentric	heliocentric
	moons	particle	stars

Galileo also used his telescope to observe the ..... of Jupiter.
His observations provided evidence to support the ..... model of the Solar System.

(iii) Describe how a reflecting telescope is different from the simple telescope which Galileo used.

(2)

.....

.....

.....

.....

Q3.

When light strikes a glass surface it can be both refracted and reflected.
In 1610 Galileo used a refracting telescope to observe the planet Jupiter.

(i) Explain how a refracting telescope produces a magnified image of Jupiter.

(3)

.....

.....

.....

.....

.....

.....

(ii) In 1610, the geocentric model of the Solar System was commonly accepted.

Explain how Galileo's observations contradicted the geocentric model.

(3)

.....

.....

.....

.....

.....

.....

Q4.

* A long time ago, astronomers thought that the Earth was the centre of the Universe.

This was called the geocentric model.
The evidence for this model came from observations of the sky using the naked eye.

After the telescope was invented, astronomers quickly gathered evidence which showed that the geocentric model is not correct.

Describe the evidence both for the geocentric model and against the geocentric model.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Q5.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 15 shows an artificial satellite in orbit around the Earth.

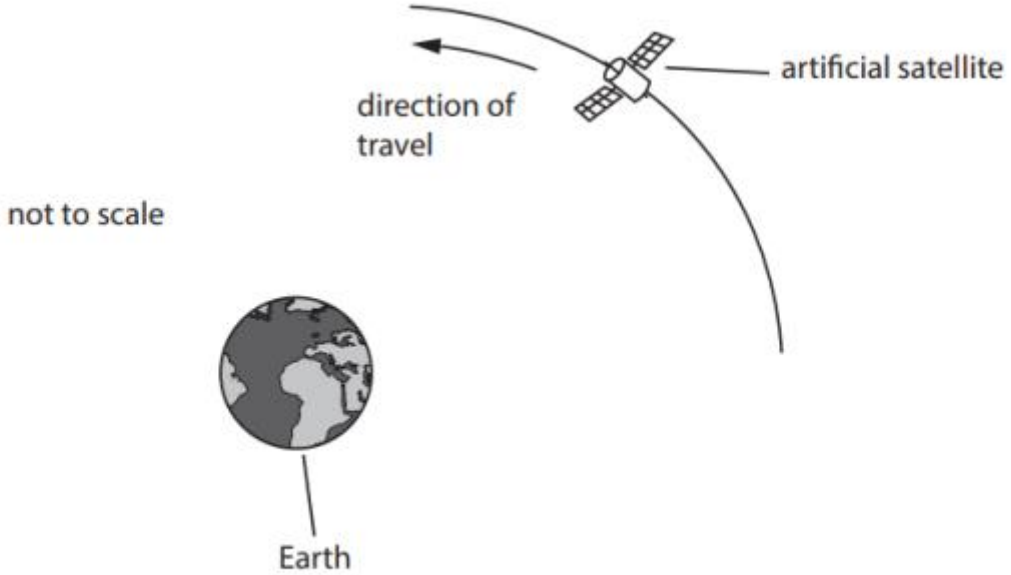
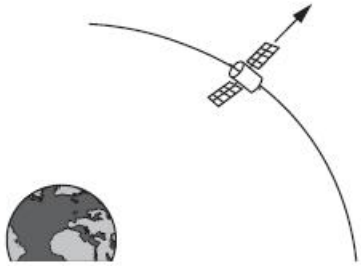


Figure 15

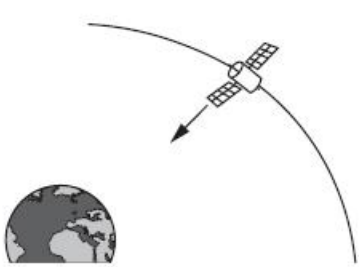
(i) Which of these shows the direction of the force that keeps the satellite in orbit around the Earth?

(1)

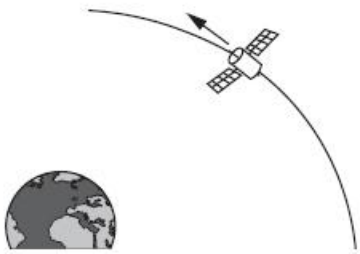
☐ A



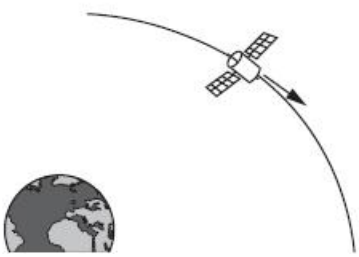
☐ B



☐ C



☐ D



(ii) The satellite travels at a constant speed around the Earth.

Explain why the satellite is accelerating even though it travels at a constant speed.

(2)

.....

.....

.....

.....

(Total for question = 3 marks)

Q6.

Which row has two correct statements about black holes?

(1)

	the gravitational field of a black hole	a black hole is formed when
☐ A	allows only electromagnetic radiation to escape	a nebula collapses
☐ B	allows nothing to escape	a very large star collapses
☐ C	allows nothing to escape	a nebula collapses
☐ D	allows only electromagnetic radiation to escape	a very large star collapses

(Total for question = 1 mark)

Q7.

Asteroids have elliptical (oval) orbits around

(1)

- ☐ **A** the Sun
- ☐ **B** the Moon
- ☐ **C** Mars
- ☐ **D** the Earth

(Total for question = 1 mark)

Q8.

The Moon orbits the Earth.

Which of these describes the Moon?

(1)

- ☐ **A** an asteroid
- ☐ **B** a comet
- ☐ **C** a nebula
- ☐ **D** a natural satellite

(Total for question = 1 mark)

Q9.

The Sun is at the centre of our Solar System.

The planets orbit the Sun.

Which planet is nearest to the Sun?

(1)

- ☐ **A** Jupiter
- ☐ **B** Mars
- ☐ **C** Mercury
- ☐ **D** Venus

(Total for question = 1 mark)

Q10.

Which of these planets is at the greatest distance from the Sun?

(1)

☐ **A** Jupiter
☐ **B** Mars
☐ **C** Neptune
☐ **D** Venus

(Total for question = 1 mark)

Q11.

(a) Put a cross (☒) in the box next to your answer.

Which of these is the biggest?

(1)

☐ **A** the Solar System

☐ **B** a galaxy

☐ **C** a nebula

☐ **D** the Universe

(b) These are four stages in the evolution of a star similar to the Sun.
They are **not** in the correct order.

1. main sequence star
2. white dwarf
3. red giant
4. nebula

Write down the stages in the correct order.

(2)

The first stage has been done for you.

..... nebula ..... .

.....

.....

.....

(c) (i) The chart shows the electromagnetic (EM) spectrum.
Some parts of the spectrum have been labelled.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays

State the name of part **Q**.

(1)

.....

(ii) Three telescopes using different parts of the spectrum have been added to the chart.

radio	**P**	**Q**	visible	**R**	**S**	gamma

			light			rays
Jodrell Bank			Hubble			Compton

Jodrell Bank is located near Manchester, whereas Hubble and Compton are in space.

Explain why some telescopes are located outside the Earth's atmosphere.

(2)

.....

.....

*(d) Describe how modern telescopes have contributed to our understanding of the Universe.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question is 12 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	gravitational / centripetal (1)	accept gravity / weight / gravitational field strength	(1) AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	arrow from the satellite towards the centre of Earth – by eye (1)	direction must be clear	(1) AO 2 1

Q2.

	Answer	Acceptable answers	Mark
(i)	D		**(1)**
(ii)	moons (1) heliocentric (1)	must be in correct order	**(2)**
(iii)	A description including two of the following points Reflecting telescope has mirror(s) (1) Galilean telescope has only lenses (1) Reflecting telescope can gather more light / can have a larger objective (1) Image viewed from the side of reflecting telescope (1) Image viewed from end of Galilean telescope. (1)	refracting telescope reverse argument	**(2)**

Q3.

	Answer	Acceptable answers	Mark
(i)	An explanation linking **three** of the following: • the (objective) lens (1) • collects light (from Jupiter) (1)	Marks can be awarded for a labelled diagram	**(3)**

	• focuses/converges/refracts the light (1) • (objective) produces a real image (1) • (at the) focal point (1) • (Image is viewed by the) eyepiece lens (1) • (Eyepiece) produces a virtual image (1)		
(ii)	An explanation linking the following: (the geocentric model) • has the Earth at the centre of the Universe/solar system / everything orbits the Earth (1)(but Galileo observed) • moons of Jupiter (1) • (so) not all things orbit the Earth/ geocentric model is wrong (1)	**Accept alternative argument:** Galileo observed moons/stars/objects Orbiting/going round Jupiter (this) suggested that not everything orbits the Earth	**(3)**

Q4.

Question Number		Indicative Content	Mark
QWC	*	A description including some of the following points evidence for • idea of Sun, Moon, stars or planets moving across the sky (not just orbiting) • in the same direction • pattern is repeated • appear to be going around the Earth • same every day evidence against • moons of {Jupiter/ other planet (with moons)} • appear to {orbit/ go around} {Jupiter/ other planet} • movement of Sun etc. not quite the same each day • planets do not move in a simple path • retrograde (west-east) motion of planets If no other marks scored • heliocentric model = Level 1	**(6)**

Level	**0**	No rewardable content
1	**1 - 2**	• a limited description stating one fact for or against e.g. for – the Sun / stars move across the sky OR against - Jupiter has moons OR against - (Galileo) produced the {heliocentric / sun-centred} model • the answer communicates ideas using simple language and uses limited scientific terminology e.g. some correct names for the moving objects • spelling, punctuation and grammar are used with limited accuracy
2	**3 - 4**	• a simple description involving (linked) facts e.g. the Sun and stars move across the sky **AND** do the same thing each day OR moons orbit Jupiter OR one fact for **AND** one against e.g. the sun moves across the sky but changes from day to day • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately e.g. correct names for the moving objects • spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed description of arguments for **AND** against, including at least one link. e.g. the Sun and stars move across the sky. Galileo observed moons, which orbit Jupiter. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q5.

Question number	Answer	Additional guidance	Mark
(i)	 B A is incorrect because the force of gravity acts towards the centre C and D are incorrect because the force does not act tangentially		**1** **AO1.1**

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking any **two** from velocity is changing (1) direction is changing (1) there is a resultant force (on the satellite) (1)	 velocity / acceleration is a vector accept unbalanced forces the direction of the velocity is changing scores 2 marks	**2** **AO1.1**

Q6.

Question number	Answer	Mark
	B	**(1)**

Q7.

Q8.

Question number	Answer	Additional guidance	Mark
	D a natural satellite A is incorrect, the Moon is not an asteroid B is incorrect, the Moon is not a comet C is incorrect, the Moon is not a nebula		(1) AO1

Q9.

Question number	Answer	Mark
	C Mercury A is incorrect Jupiter is the fifth planet from the Sun B is incorrect Mars is the fourth planet from the Sun D is incorrect Venus is the second planet from the Sun	(1) AO1

Q10.

Question number	Answer	Mark
	☒ C Neptune	(1)

Q11.

	Answer	Acceptable answers	Mark
(a)	D the Universe (1)		**(1)**
(b)	(nebula) main sequence (star) (1) AND red giant white dwarf (1) All three in correct order for 2 marks	Red Giant White Dwarf (Main sequence) (1)	**(2)**
(c)i	infrared (radiation)/(rays) (1)		**(1)**
(c)ii	An explanation linking any two from • above the clouds / no clouds/ no weather (1) • image is clearer/more detailed/ not distorted/not blurred (1) • no light pollution (1) • (some) telescopes use gamma/ X-rays/ultraviolet /infrared/microwaves (1) • no absorption (by atmosphere) of gamma/ Xrays/ ultraviolet /infrared/ microwaves (1)	Credit to be given for stating that all telescopes would be better in space, but size and weight may exclude e.g. Jodrell Bank from space. no air/dust/pollution wider field of view/ can use anytime IG NORE 'see further' IGNORE 'it is closer (to the stars/planets)' IGNORE: references to improving understanding / knowledge of space	**(2)**

		Indicative Content	Mark
QWC	***(d)**	A description including some of the following points • improved QUALITY eg higher or better magnification/ detail/resolution or clearer/brighter image **OR** MORE INFORMATION (than with naked eye) of image/data eg new planets/stars/nebula e/pulsars (This could be extra detail for greater	**(6)** **Exp**

		magnification/resolution only) • detection of (non-visible) electromagnetic WAVES eg Xray / UV / IR/ radio • TECHNOLOGY that enable collection of more data eg reflecting telescope/arrays ***and/or*** additions eg computer-aided /photographic connections or larger (objective) lens/mirror • POSITION of telescopes – eg orbital/outside atmosphere/on top of mountains/away from atmosphere/rays not absorbed/obscured/scattered by atmosphere. Ignore 'Hubble' or 'Compton'.	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited description e.g. mention of any one example such as "magnifies stars/planets" OR "discovering new planets/stars" • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple description e.g. mention of either two of the improvements **OR** extra detail about one of the improvements eg improvement **plus** example (ie Magnifies planets *so that craters/mountains may be seen)* • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed description e.g. mention of three (or more) improvements **OR** two improvements **plus extra detail about one**	

		of them (ie Telescopes in space can detect XRays *that would be absorbed by the atmosphere*) • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

(Suitable extra detail shown in italics in examples above)

Total for Question = 12 marks